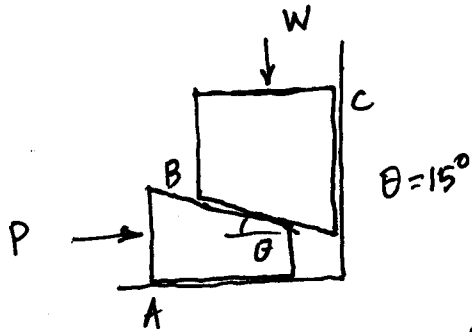


Two Wedge Example Problem : Raising Case

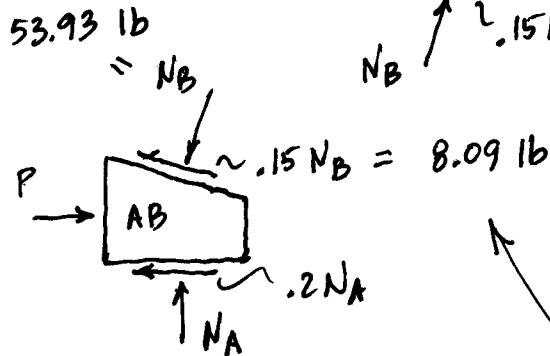


let $W = 50 \text{ lb}$.

$\mu_A = .2$, $\mu_B = .15$, $\mu_C = 0$

Wedges are massless. Find P to raise wedge BC.

FBD's:



Draw μN frictions to oppose relative motion

$$\underline{BC} + \uparrow N_B \cos \theta - .15 N_B \sin \theta - 50 = 0$$

$$50 = N_B (\cos 15 - .15 \sin 15)$$

$$= N_B (.9271)$$

$$\boxed{N_B = 53.93 \text{ lb}}$$

$$\boxed{.15 N_B = 8.09 \text{ lb}}$$

$$\underline{AB} + \uparrow N_A - 53.93 \cos 15 + 8.09 \sin 15 = 0$$

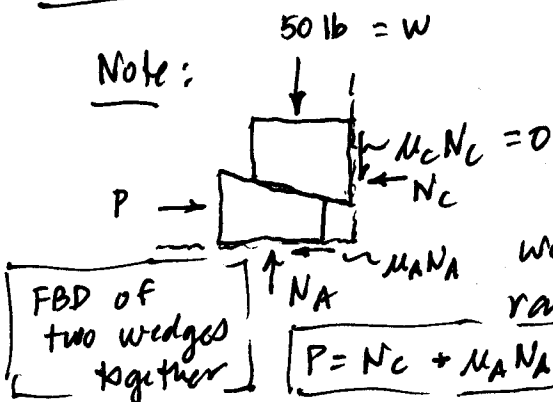
$$\boxed{N_A = 50 \text{ lb}} \quad (\text{as expected})$$

$$\boxed{.2 N_A = 10 \text{ lb}}$$

$$\rightarrow P - 10 - 53.93 \sin 15 + 8.09 \cos 15 = 0$$

$$\boxed{P = 31.77 \text{ lb}} \quad \text{to raise wedge BC.}$$

Note:



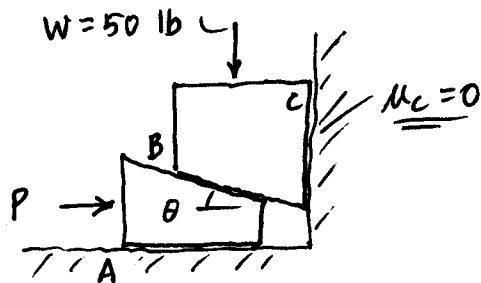
$N_A = 50 \text{ lb}$ because $\mu_C = 0$ at wall.
 $= W$

If μ_C was non-zero, then $\mu_C N_C$ would act vertically (down for the raising case), and $\boxed{N_A = 50 + \mu_C N_C}$

$$\boxed{P = N_C + \mu_A N_A}$$

Two-Wedge Problem : Slightly different approach

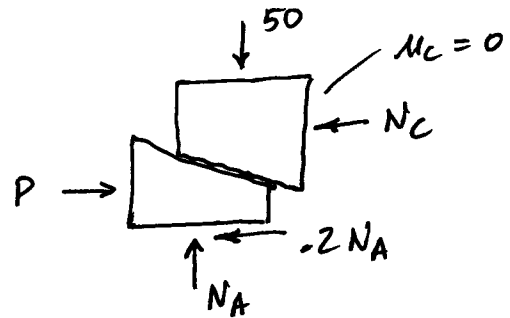
This method requires that the wall is smooth.



$$\theta = 15^\circ, \quad W = 50 \text{ lb}$$

$$\mu_C = 0; \quad \mu_A = .2; \quad \mu_B = .15$$

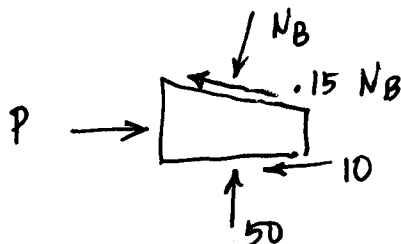
Start w/ two wedge FBD:



Since $\mu_C = 0$, $+\uparrow$

$N_A = 50 \text{ lb}$
$.2N_A = 10 \text{ lb}$

Then draw FBD of lower Wedge:



$$+\uparrow \quad 50 - N_B \cos 15 + .15 N_B \sin 15 = 0$$

$$N_B = \frac{50}{\cos 15 - .15 \sin 15}$$

$N_B = 53.93 \text{ lb}$
$.15 N_B = 8.09 \text{ lb}$

$$\rightarrow P - 10 - 8.09 \cos 15 - 53.93 \sin 15 = 0$$

$P = 31.77 \text{ lb} \leftarrow$

Raising
Case